

FULLY SOLVED SAMPLE PAPER UNDERGRADUATE STUDIES ADMISSION TEST (USAT-M)

Having 12 years (Intermediate or Equivalent Education)

PART-I:
Verbal Reasoning 20 MCQs
Quantitative Reasoning 25 MCQs
USAT-M Subjects: 30 MCQs

Time: 100 Minutes
Marks: 75 Marks

PART-II:
Essay Writing Skills 25 Marks

Time: 40 Minutes

Instructions: Please read the following instructions carefully before attempting the paper.
75 MCQs are given on various topics, which carry equal marks. Each correct answer carries 1 mark.

- Attempt all the questions.
- In every question, four / five options are given (A, B, C, D, E), you have to select only one correct option.
- Mark your choice by filling in the appropriate circle against each question.
- Use ball point pen (black or blue) to shade / blacken the corresponding circle in the answer sheet.
- The candidates should carefully think about their answer before filling the circles on the answer sheet.
- Erasing, cutting or overwriting is not allowed. Once an answer has been given on the answer sheet, the candidate will not be permitted to change any of his/her answer in any way. All such answers will be treated as wrong.
- The candidate should not write any thing regarding answers on the question paper. All answers must be given on the answer sheet only.

PART-I

A: VERBAL REASONING

ANALOGY

- Jesus is related to Christians in the same way as Zoroaster is related to?
(a) Jews (b) Parsis✓
(c) Tribals (d) Catholics
- South is related to North-West in the same way as West is related to?
(a) South-West (b) East
(c) North-East✓ (d) South
- Summit is related to Apex in the same way as Summon is related to?
(a) Court (b) Judge
(c) Witness (d) Beckon✓
- Distil is related to Whisky in the same

way as Brew is related to?

- Ferment (b) Gin
 - Beer✓ (d) Sugar
- DDT is related to Abbreviation in the same way as LASER is related to?
(a) Antithesis (b) Acronym✓
(c) Epigram (d) Epithet

A2: SYNONYM

6. **STRIDENCY:**

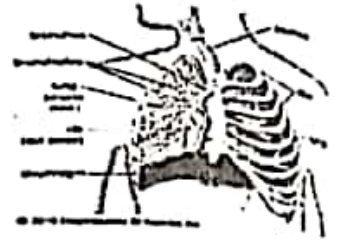
- Harshness✓ (b) Flippant
- Consistency (d) Stress

7. **ACUTE:**

- Curious (b) Severe✓

70. In Down's syndrome, which one of the following pair of chromosome fails to segregate?
A) 7 B) 21✓ C) 18 D) 19

71. Label the part 'Y' in the following diagram:
A) Pleura
B) Chest cavity
C) Diaphragm✓
D) Intercoastal muscles



72. During photorespiration, the glycolate is converted into glycine in a structure of cell called:
A) Golgi Bodies B) Mitochondria C) Glyoxisome D) Peroxisome✓

73. The reaction between the phosphate group of one nucleotide and hydroxyl group of another is a _____ synthesis in DNA molecule.
A) Dehydration✓ B) Oxidation C) Rehydration D) Reduction

74. Enzyme which attaches the Okazaki fragments in lagging strand is called:
A) Restriction endonuclease B) DNA helicase C) Primase D) DNA ligase✓

75. Haemoglobin exhibits:
A) Secondary Structure B) Quaternary Structure✓
C) Primary Structure D) Tertiary Structure

PART-II

NOTE: (i) PART-II is to be attempted on separate Answer Book.

(ii) Attempt all questions from PART-II.

(iii) Extra attempt of any question or any part of the attempted question will not be considered.

(iv) Candidate must write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.

Write a Argument Based Essay on any one of the following:

- i) Patriotism
- ii) Indiscipline Among Students
- iii) Is Science a Curse or A Blessing?

Write a Narrative Based Essay on any one of the following:

- i) A Journey By Train
- ii) My First Day At School
- iii) My Hobby

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- (c) Neither (A) nor (B) is true (d) Both (A) and (B) are true✓
19. "In literature as in finance... much paper and much poverty may co-exist." What does it mean?
 (a) Jam Sahib was rich man as he was the king of Nawanagar (b) Shrewsbury was a poor man
 (c) The cricketer who gets lot of runs may not play an enjoyable cricket✓ (d) None of these
20. What gives cricket its character?
 (a) The spirit of sacrifice (b) The spirit of joyous adventure
 (c) The gaiety daring attitude of the player (d) All of these✓

B: QUANTITATIVE REASONING

BI: ARITHMETIC

21. The chairs in the school hall can be set out in 35 equal rows or in 45 equal rows or in 105 equal rows are:
 (a) 600✓ (b) 400
 (c) 40 (d) 80
22. Three bells toll after intervals of 6, 9 and 15 minutes, respectively. If they toll together at 5 p.m., when will they toll together next?
 (a) 6:30✓ (b) 5:30
 (c) 6:45 (d) 5:45
23. It takes Riaz 30 minutes to mark a paper. Razi only need 25 minutes to mark a paper. If they both start marking papers at 11:00 AM, what is the first time they will finish marking a paper at the same time?

- a) 12:30 b) 12:45
 c) 1:30✓ d) 12:25
24. Sonia buys two off-cuts of ribbon in a sale. One is 153 cm long. The other is 204 cm long. She cuts them so that she ends up with a number of pieces all the same length. What is the greatest length each piece can be?
 (a) 39 (b) 6
 (c) 17 (d) 51✓
25. A farmer wants to fence a triangular field. He plans to put a fencing post in each corner and place other posts at equal distance along its sides. He wants the posts to be as far apart as possible. The sides of the field are 477 feet 2412 feet and 639 feet long. How far apart will the posts be?
 (a) 18 feet (b) 9 feet✓
 (c) 27 feet (d) 159 feet
26. Which of the following is greater than p when $p = \frac{7}{11}$?
 I. $\frac{1}{p}$ II. $\frac{p+1}{p}$ III. $\frac{p+1}{p-1}$
 (a) only I (b) I and II only✓
 (c) I and III only (d) II and III only

B2: ALGEBRA AND FUNCTIONS

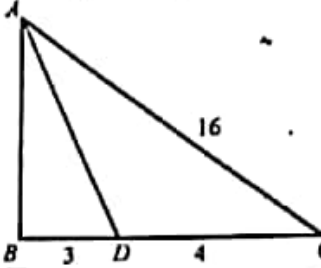
27. A solid cylinder of brass 12cm high and 6cm in radius is melted and recasted into a right circular cone of diameter 16cm. Find the height of the cone.
 (a) 10.15cm (b) 20.25cm✓
 (c) 28.75cm (d) 26.28cm
28. The slant height l of a cone with radius of base r and height h is given by $l =$
 (a) $\sqrt{h^2 + r^2}$ ✓
 (b) $\sqrt{h^2 - r^2}$
 (c) $\pi r \sqrt{h^2 + r^2}$
 (d) $\pi r \sqrt{h^2 - r^2}$
29. If a , b and c are the measures of the sides of a triangle, then area of triangle is
 (a) $\sqrt{s(s+a)(s+b)(s+c)}$, where $2s = a+b+c$

- (b) $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$ ✓
 (c) $\sqrt{(s-a)(s-b)(s-c)}$, where $3s = a+b+c$ ✓
 (d) $\sqrt{s(s-a)(s-b)(s-c)}$, where $2s = a+b+c$ ✓

30. Which of these sets is not a null set?
 (a) $A = \{x \mid 6x = 24 \text{ and } 3x = 1\}$
 (b) $B = \{x \mid x + 10 = 10\}$ ✓
 (c) $C = \{x \mid x \text{ is a man older than 200 years}\}$
 (d) $D = \{x \mid x < x\}$

B3: GEOMETRY

31. $20^\circ =$
 (a) $360'$ (b) $630'$
 (c) $1200'$ ✓ (d) $3600'$
 32. $\frac{3\pi}{4}$ radians =
 (a) 115° (b) 135°
 (c) 150° ✓ (d) 30°
 33. In the following triangle, $AD =$



- (A) $3\sqrt{2}$ (B) $6\sqrt{6}$ ✓
 (C) $6\sqrt{3}$ (D) $3\sqrt{7}$

EQUATIONS

1. If $3x + 17 = 9 - x$, what is the value of x ?
 (A) 2 (B) 3
 (C) -2 ✓ (D) -3
 2. If $x - 5 = 9$, what is the value of $x^2 - 5$?
 (A) 196 (B) 191 ✓
 (C) 16 (D) 11
 3. If $at - b = c - dt$, what is the value of t in terms of a, b, c and d ?
 (A) $\frac{b-c}{a-d}$ (B) $\frac{a}{b}$

- (C) $\frac{c}{d}$ (D) $\frac{b+c}{a+d}$ ✓

B5: STATISTICS

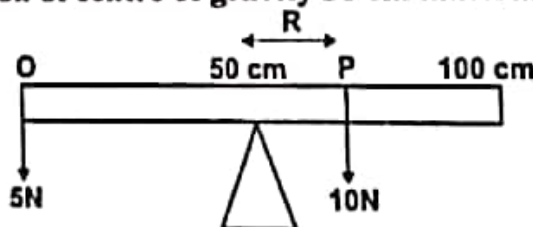
37. Primary data and ungrouped data are:
 A Same B Opposite
 C Not same ✓ D Proportional
 38. Statistical laws are true:
 A On the average ✓ B In each case
 C In long run D None of these
 39. The data which have not undergone any statistical treatment are _____ data.
 A Primary ✓ B Secondary
 C Grouped D None of these

B6: SCENARIO BASED/MENTAL MATHEMATICS

40. A tank 30 cm by 20 cm by 10 cm is $\frac{1}{5}$ full of water. How much water is in the tank?
 (A) 3000 cm^3 (B) 6000 cm^3
 (C) 1200 cm^3 ✓ (D) 1000 cm^3
 41. During the first year, the population of a town increased by 4% and during the second year it diminished by 4%. If at the end of 2nd year, its population, was 2596, in the beginning it was:
 (A) 2,550 (B) 2,600
 (C) 2,400 (D) $2,500$ ✓
 42. A solution of 27 gallons of acid contains 9 gallons of pure acid. How much water, in gallons, should be added to produce a 25 per cent solution of this acid?
 (A) 9 ✓ (B) 18
 (C) 27 (D) 25
 43. At p paisa per orange, what is the price, in rupees, for 1 dozen oranges?
 (A) $\frac{p}{12}$ (B) $\frac{p}{100}$
 (C) $\frac{12p}{100}$ (D) $\frac{12p}{100}$ ✓
 44. Which of the following numbers belongs to the series given below: 7, 18, 29, 40,
 (A) 2,095 (B) 2,007
 (C) 2,094 (D) $2,097$ ✓
 45. In a camp, there is provision for 1,600 participants to last 60 days. If the present strength of the camp is 1,200, the provision will last for _____ days.
 (A) 96 (B) 86
 (C) 80 ✓ (D) 72

PHYSICS

46. An electric charge in uniform motion produces:
 A) An electric field.
 B) Both magnetic and electric fields. ✓
 C) A magnetic field.
 D) Neither magnetic nor electric fields.
47. If 'm' is the mass, 'c' is the velocity of light and $x = mc^2$, then dimensions of 'x' will be:
 A) $[LT^{-1}]$
 B) $[MLT^{-1}]$
 C) $[ML^2T^{-2}]$ ✓
 D) $[MLT^{-2}]$
48. For a body to be in complete equilibrium
 A) Linear acceleration is zero
 B) Angular acceleration is zero
 C) Linear acceleration is zero but angular acceleration is not zero
 D) Linear acceleration and angular acceleration both should be zero ✓
49. If a force of magnitude 8 N acts on a body in direction making an angle 30, its x and y
 A) $F_x = 4\sqrt{3}$ and $F_y = 8$
 B) $F_x = 4\sqrt{3}$ and $F_y = 4$ ✓
 C) $F_x = 8$ and $F_y = 4\sqrt{3}$
 D) $F_x = 8\sqrt{3}$ and $F_y = 4$
50. The difference of a vector $\vec{B}$ and its negative vector $-\vec{B}$ is
 A) A null vector
 B) Equal to magnitude of vector $\vec{B}$
 C) Twice the magnitude of vector $\vec{B}$ ✓
 D) Smaller than magnitude of vector $\vec{B}$
51. Time of projectile's flight is
 A) $\frac{v_i^2 \sin^2 \theta}{g}$
 B) $\frac{v_i^2 \sin \theta}{g}$
 C) $\frac{2v_i \sin \theta}{g}$ ✓
 D) $\frac{v_i^2}{g} \sin \theta$
52. Two forces, 5 N and 10 N are acting at 'O' and 'P' respectively on a uniform meter rod suspended at the position of centre of gravity 50 cm mark as shown in the figure.



What is the position of 'P' on meter rod?

- A) 80 cm
 B) 70 cm
 C) 75 cm ✓
 D) 65 cm
53. In Millikan's Method, the radius of droplet can be calculated by:
 A) $r = \sqrt{\frac{qv_t}{2pg}}$
 B) $r^2 = \frac{9\eta v_t}{2pg}$ ✓
 C) $r^2 = \frac{q\eta v_t}{pg}$
 D) $r = \frac{9\eta v_t}{2pg}$

CHEMISTRY

54. Which one of the following has same number of molecules as present in 11g of C_2H_2 purification he had made 12g of product. Which of the following is percentage yield:
 A) CH_2O
 B) C_3H_6O
 C) $C_4H_{12}O_2$ ✓
 D) CH_4O
55. The stability in the following structure is due to the:
 A) Disulfide bridges
 B) Hydrogen bonding between NH ✓
 C) presence of unpaired electron in the structure
 D) weak vander Waals force

- (c) Accidental (d) Rice
- ANTONYME**
8. **OFTEN:**
 (a) Quickly (b) Never
 (c) Never (d) Sometimes✓
9. **REJECT:**
 (a) Allow (b) Agree
 (c) Refuse (d) Accept✓

SENTENCE COMPLETION

10. Stop taking drugs lest you are caught.
 (A) Might be caught✓
 (B) Will be caught
 (C) May be caught
 (D) Would be caught
11. It is *all and one* to me whether he lives in Karachi or Hyderabad.
 (A) All but one
 (B) All one
 (C) One and the same✓
 (D) All or one
12. In his lecture, he dealt *about* the cause of the Gulf War.
 (A) With✓ (B) On
 (C) For (D) No improvement
13. It is *one and quarter* hours since Haris went away.
 (A) One and quarter
 (B) One and quarter hour
 (C) One hours and quarter
 (D) One hour and a quarter✓
14. Where politics fails, economics *may* sometime succeed.
 (A) May sometimes succeeds✓
 (B) May sometime succeeds
 (C) Sometimes succeed
 (D) Sometimes succeeds
15. He has been working *off and on* for several years to compile a dictionary.
 (A) Regularly (B) Constantly
 (C) On and off✓ (D) On or off

COMPREHENSION

PASSAGE

"But we do not judge a cricketer so much by the runs he gets as by the way he gets them." In literature as in finance, says Washington Irving, "much paper and much poverty may co-exist." And in cricket, too many runs and much

dullness may be associated. If cricket is menaced with creeping paralysis, it is because it is losing the spirit of joyous adventure and becoming a mere instrument for compiling tables of averages. There are dull, mechanical fellows who turn out runs with as little emotion as a machine turns out pins. There is no colour, no enthusiasm, no character in their play. Cricket is not an adventure to them, it is a business. It was so with Shrewsbury. His technical perfection was astonishing; but the soul of the game was wanting in him. There was no sunshine in his play, no swift surprise of spending unselfishness. And without these things, without gaiety, daring and the spirit of sacrifice cricket is a dead thing. Now the Jam Sahib has the root of the matter in him. His play is as sonny as his face. He is not a miser hoarding up runs, but a millionaire spending them, with a splendid yet judicious prodigality. It is as though his pockets are bursting with runs that he wants to shower with his blessing upon the expectant multitude. It is not difficult to believe that in his little kingdom of Nawanagar where he has the power of life and death in his hands, he is extremely popular, for it is obvious that his pleasure is in giving pleasure.

16. What is/are true of Shrewsbury?
 I. He was a cricket player.
 II. His technical knowledge about cricket was poor.
 III. There was no spirit in his play.
 (a) I and II (b) I and III✓
 (c) Only I (d) None of these
17. The author feels that:
 (a) Technical perfection is not required in playing an enjoyable cricket (b) He who pays cricket with adventure and enthusiasm makes it enjoyable✓
 (c) He who scores a century must be a good cricketer (d) Cricket is a monotonous game
18. Jam Sahib:
 (a) Was a splendid cricket player (b) Lived in Nawanagar

- group of one peptide and CO group of another peptide
56. While finding the relative atomic mass, which of the following standard is used to compare the atomic mass of chlorine (35.5 amu):
 A) Carbon - 12✓ C) Carbon - 13
 B) Neon - 20 D) Nucleon number
57. Liquid in the container have temperature 70°C. What will be temperature in Kelvin scale?
 A) 283 K C) 343 K✓
 B) 350 K D) 300 K
58. Number of neutrons in $^{66}_{30}\text{Zn}$ will be:
 A) 30 B) 38 C) 35 D) 36✓
59. The maximum number of electrons in electronic configuration can be calculated by using formula:
 A) $2l + 1$ B) $2n^2$ ✓ C) $2n^2 + 2$ D) $2n^2 + 1$
60. Isotopic symbol of ion of sulphur -33 is $^{33}_{16}\text{S}^{-2}$. How many no of protons and neutrons are present if the number of electrons are 18:
 A) $P = 18, n = 15$ B) $p = 16, n = 16$ C) $p = 16, n = 17$ ✓ D) $p = 17, n = 16$
61. Which of the following is the correct equation to calculate relative molecular mass of a gas:
 A) $M = mRT/PV$ ✓ C) $M = mPR/VT$
 B) $M = PV/mRT$ D) $M = mPRT/V$

BIOLOGY

62. The use of living organisms in industry for the production of useful products is known as
 A) Parasitology B) Biotechnology✓
 C) Biochemistry D) Molecular Biology
63. Plants having foreign DNA incorporated into their cells are called:
 A) Clone plants B) Parthenocarpic plants
 C) Transgenic plants✓ D) Mutant giants
64. Treatment by using attenuated culture of bacteria is called
 A) Chemotherapy B) Antisepsis
 C) Sterilization D) Vaccination✓
65. Label 'a' in the following diagram:
 A) Cardiac sphincter B) Stomach valve
 C) Sinoatrial valve D) Pyloric sphincter
66. The major cause of hepatitis B is
 A) Blood transfusion✓ B) Absence of fibrinogen
 C) Blood clotting D) Contaminated soil
67. Internal program of events and sequences of morphological changes by which cell commit a suicide is collectively called:
 A) Necrosis B) Metastasis C) Epistasis D) Apoptosis✓
68. Phragmoplast is formed from vesicle which originates from:
 A) Smooth Endoplasmic Reticulum B) Ribosome
 C) Golgi Complex✓ D) Rough Endoplasmic Reticulum
69. A group of ribosomes attached to messenger RNA is known as:
 A) Ribosome B) Nucleosome C) Lysosome D) Polysome✓

